

ProgHard Link

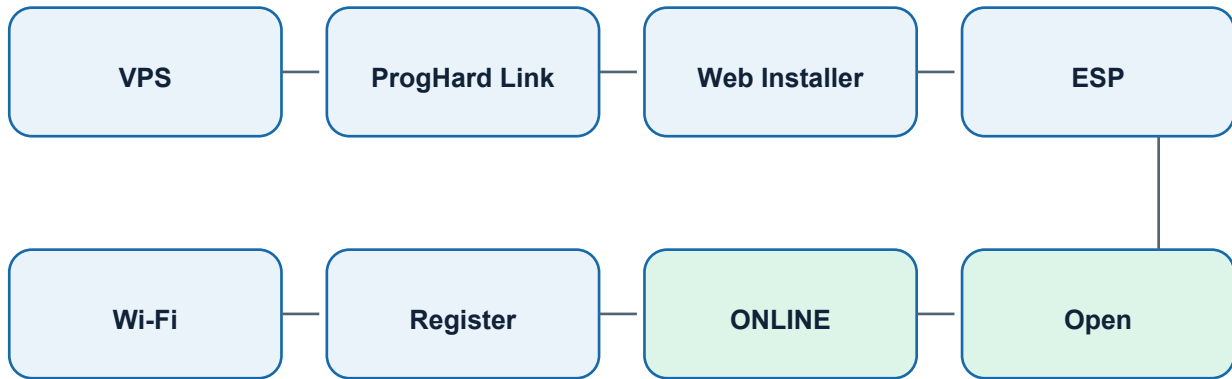
Getting Started Guide

Deploy your own instance, flash an ESP, and open its interface remotely without forwarding a port to the device.

Intermediate edition - installation and remote access

Server 0.5.0 · Base 0.2.3 · Arduino library 0.4.8 · espway-tunnel/2

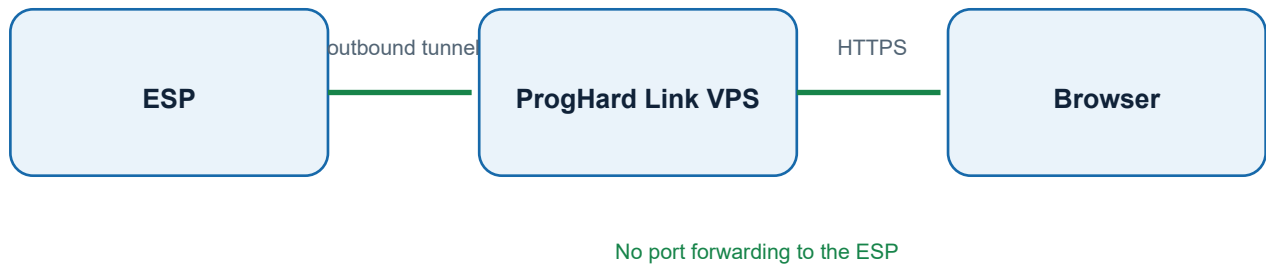
ProgHard Link in 10 minutes



The first VPS deployment naturally takes longer than ten minutes. Once the instance is ready, this path summarizes device installation.

1. Understand the architecture

ProgHard Link is a self-hosted, local-first stack. The ESP owns its HTTP interface on the LAN. For remote access, it opens an outbound WebSocket tunnel to your VPS.



- The browser uses HTTPS to Caddy on the VPS.
- The server maps the request to the device tunnel stream.
- No port forwarding to the ESP is required.
- The local interface remains usable without Internet access.

2. Requirements and versions

This guide is for hobbyists comfortable with a terminal and DNS. Arduino and OTA are not included yet.

- A current Ubuntu VPS with sudo access
- A domain and permission to create DNS records
- Docker Engine and the Docker Compose plugin
- A sacrificial ESP8266, ESP32, or ESP32-C3
- A USB data cable and Chrome or Edge

Component	Published version
Server	0.5.0
ProgHard Link Base	0.2.3
Arduino library	0.4.8
Tunnel protocol	espway-tunnel/2

3. Install the VPS

Use the official public repository. The commands below follow its minimal path; read the operator documentation before long-term public exposure.

```
git clone https://github.com/pdeprost-dot/ProgHard-Link.git /opt/proghard-link
cd /opt/proghard-link
cp .env.example .env
chmod 600 .env
# Edit .env, then run:
sudo docker compose config
sudo docker compose build
sudo docker compose pull
sudo docker compose up -d
sudo docker compose ps
```

Never run docker compose down -v on a real instance: it deletes persistent volumes.

```
ESPWAY_DOMAIN=link.example.com
ACME_EMAIL=admin@example.net
ESPWAY_OTA_OPERATOR_TOKEN=
```

4. DNS, HTTPS, and the first administrator

Create one A record for the apex and one wildcard record pointing to the VPS. The wildcard does not cover the apex.

```
link.example.com → VPS_IPV4
*.link.example.com → VPS_IPV4
admin.link.example.com → Device Manager
install.link.example.com → Web Installer
tunnel.link.example.com → outbound tunnel
esp-xxxxxx.link.example.com → device access
```

Then create the first administrator. The password is requested twice with masked input: never place it on the command line.

```
cd /opt/proghard-link
sudo docker compose exec server npm run create-admin -- --username admin
```

5. Install ProgHard Link Base

Open <https://install.link.example.com/> in Chrome or Edge, connect the ESP with a data cable, and choose Install ProgHard Link.

- Select the ESP serial port.
- Confirm ProgHard Link Base 0.2.3 and device erasure.
- Keep the page visible while the firmware is written.
- When installation completes, continue with Wi-Fi provisioning.

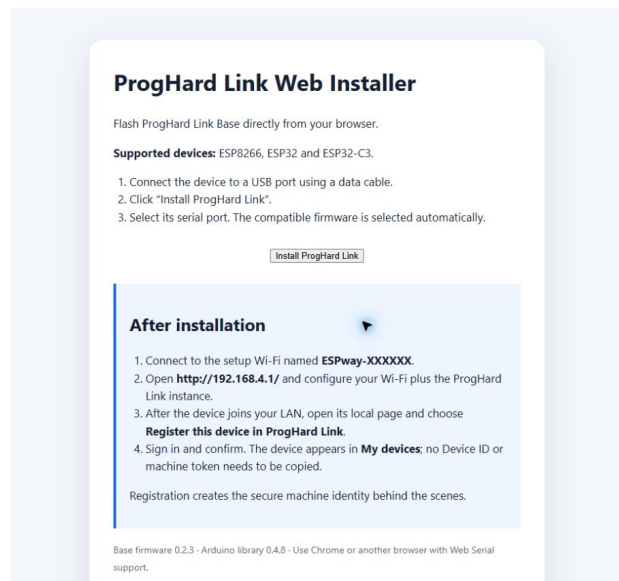


Figure 1. Web Installer home page and supported platforms.

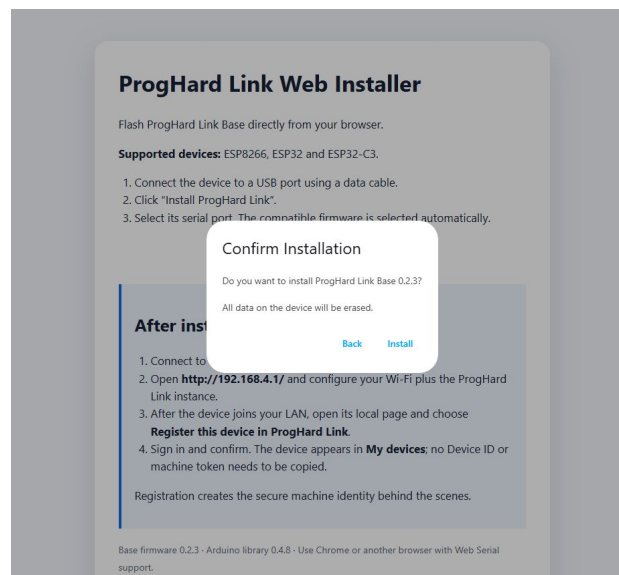


Figure 2. Confirmation for installing Base 0.2.3.

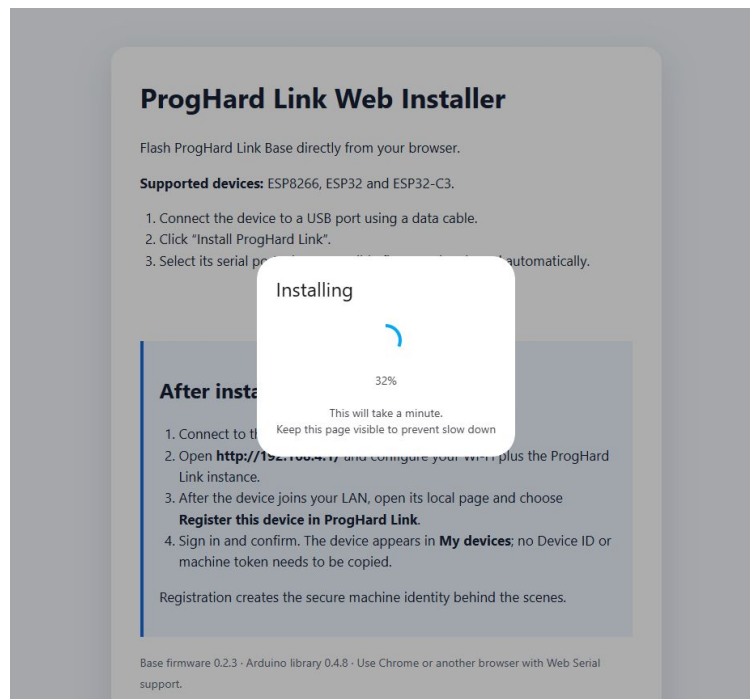


Figure 3. Real browser-based firmware write in progress.

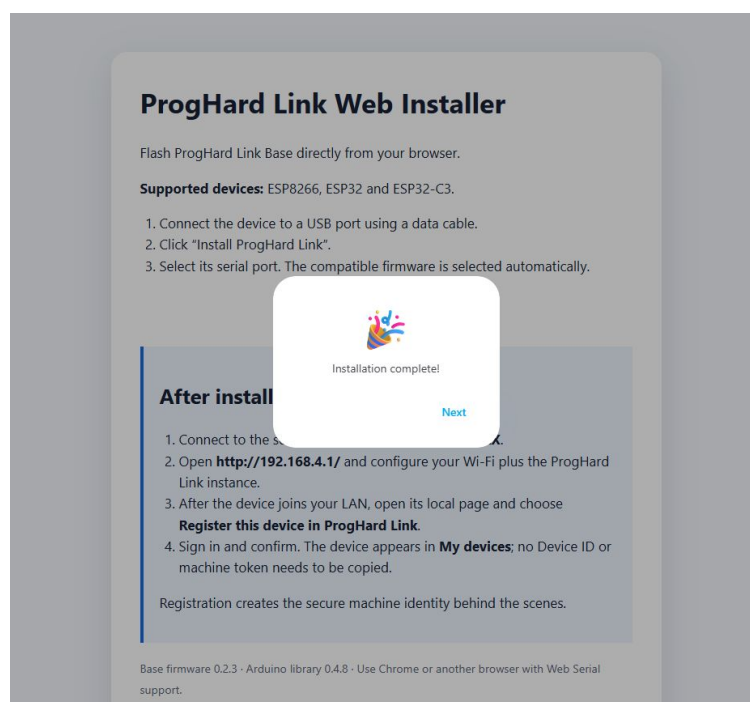
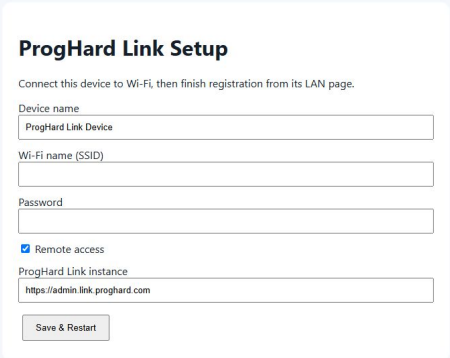


Figure 4. Installation completed.

6. Provision Wi-Fi

After restart, join the ESPway-XXXXXX access point. The captive portal should open; otherwise visit <http://192.168.4.1/>.

- Give the device a clear name.
- Enter the Wi-Fi SSID and password.
- Leave Remote access enabled if you want remote access.
- Enter the complete instance URL, for example <https://link.example.com>.
- Choose Save & Restart, then rejoin your LAN.



The screenshot shows a web form titled "ProgHard Link Setup". Below the title is a instruction: "Connect this device to Wi-Fi, then finish registration from its LAN page." The form contains several input fields: "Device name" with the value "ProgHard Link Device", "Wi-Fi name (SSID)" which is empty, and "Password" which is empty. There is a checkbox for "Remote access" which is checked. Below this is a field for "ProgHard Link instance" with the value "https://admin.link.proghard.com". At the bottom of the form is a button labeled "Save & Restart".

Figure 5. Provisioning form before entering Wi-Fi secrets.

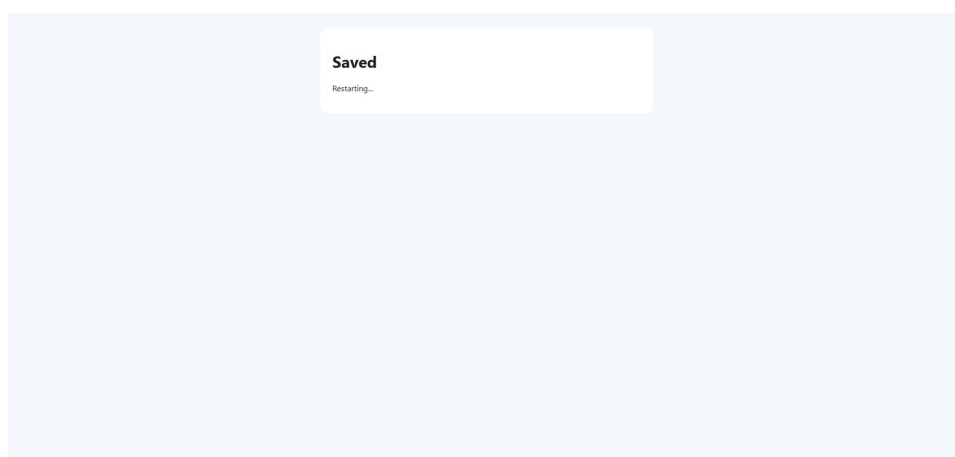


Figure 6. Configuration saved and ESP restarting.

[Device](#) [Base app](#) [Configuration](#) [Advanced OTA](#) [Diagnostics](#) [Device Manager](#)

ProgHard Link Device

Device: ProgHard Link Device

Device ID: esp-16a832

ProgHard Link

Instance: <https://admin.link.proghard.com>

Register this device in ProgHard Link

```
{
  "deviceName": "ProgHard Link Device",
  "deviceId": "esp-16a832",
  "wifi": {
    "connected": true,
    "ip": "192.168.50.234"
  },
  "remote": {
    "connected": false,
    "transport": "ws-hmac",
    "tunnelProtocol": "espway-tunnel/2"
  },
  "firmwareVersion": "0.2.3"
}
```

[Configuration](#) | [OTA](#) | [Diagnostics](#)

Figure 7. LAN page before enrollment: Wi-Fi connected, tunnel not connected yet.

7. Register the device

On the LAN, try `http://esp-xxxxxx/` first. The local page reports Wi-Fi and tunnel state. Choose Register this device in ProgHard Link.

- Sign in to Device Manager if required.
- Check the name, hardware, application, and Device ID.
- Confirm Register device.
- The success page confirms association with the account.

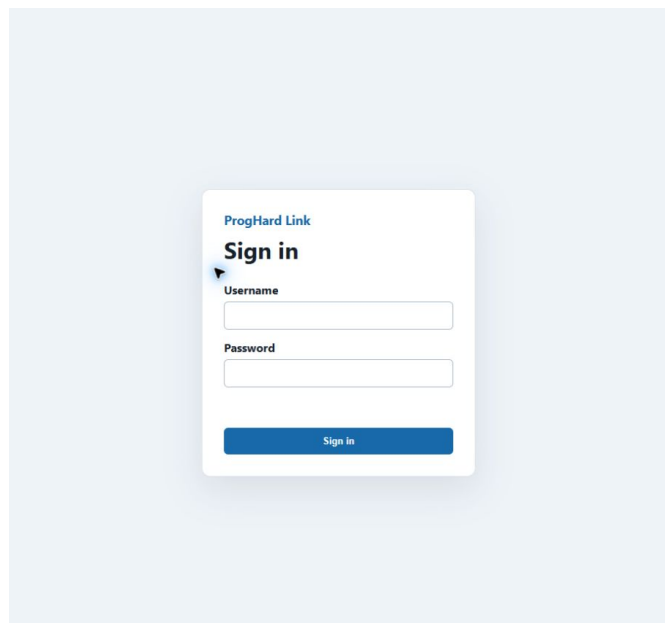


Figure 8. Human authentication for Device Manager.

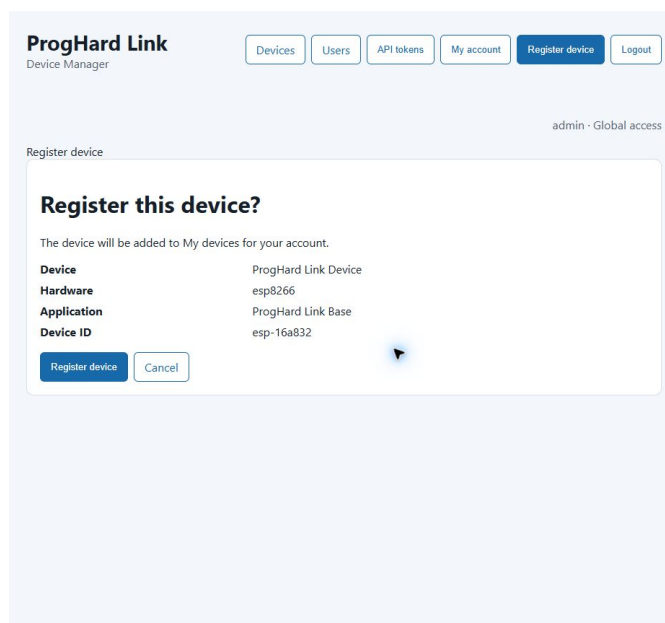


Figure 9. Confirmation of the device to register.

Device registered successfully

Device registered successfully

ProgHard Link Device is now associated with your account.

[Open device](#)[My devices](#)

Figure 10. Successful association with the account.

8. Confirm ONLINE in Device Manager

Return to My devices. Exactly one card should be present. ONLINE state and the Open button confirm that the tunnel is available.

The screenshot displays the 'ProgHard Link Device Manager' interface. At the top, the header includes the 'ProgHard Link' logo and the text 'Device Manager'. A navigation bar contains buttons for 'Devices', 'Users', 'API tokens', 'My account', 'Register device', and 'Logout'. The user is logged in as 'admin' with 'Global access'. Below this, a summary shows '1 devices · 1 online'. A single device card is displayed with the title 'ProgHard Link Device' and a green 'ONLINE' status indicator. The device ID is 'esp-16a832', the firmware is '0.2.3 · Up to date', the Wi-Fi signal is '-54 dBm (Excellent)', and the last telemetry update was '6 s ago'. At the bottom of the card are 'Open' and 'Details' buttons.

ProgHard Link
Device Manager

Devices Users API tokens My account Register device Logout

admin · Global access

1 devices · 1 online

ProgHard Link Device **ONLINE**

esp-16a832

Firmware 0.2.3 · Up to date

Wi-Fi: -54 dBm (Excellent)

Telemetry: 6 s ago

Open Details

Figure 11. Device ONLINE, firmware current, and Open available.

9. Open the remote interface

Always use My devices → Open. Device Manager creates a short-lived, single-use access ticket before the browser reaches the device HTTPS hostname.

Opening the wildcard device URL directly without authorization may return 401 authentication required. This is expected: return to Device Manager and use Open.

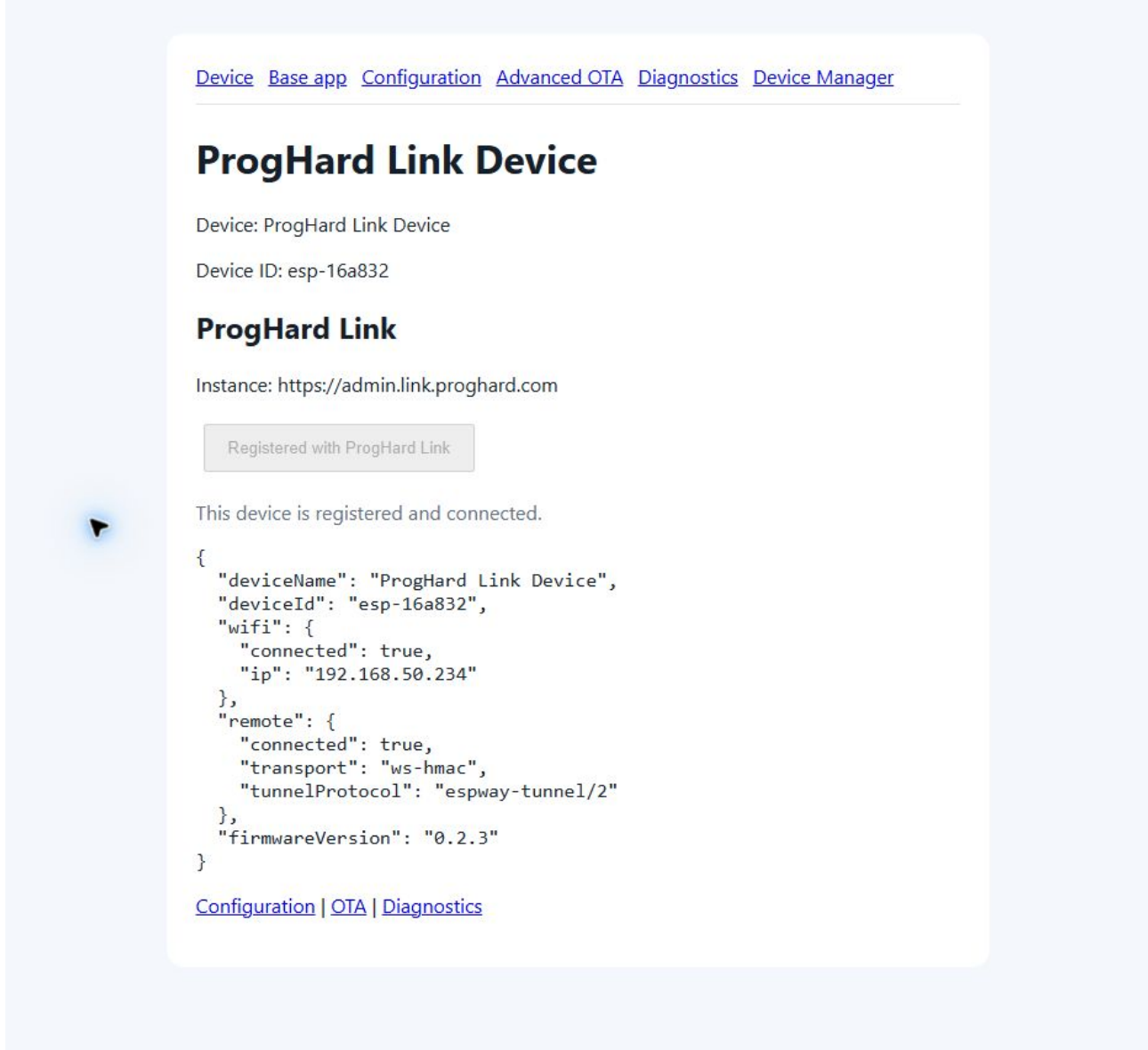


Figure 12. Real remote interface reached through My devices → Open.

10. Basic troubleshooting

The portal does not open

Open <http://192.168.4.1/> after joining ESPway-XXXXXX.

The device remains OFFLINE

Check Wi-Fi, the instance URL, DNS for tunnel.link.example.com, and system time.

device_already_registered

An authorization already exists for this Device ID. Review the old association and delete it only when the new enrollment is intentional.

401 on the device URL

Return to My devices and use Open to obtain a valid ticket.

TLS or DNS fails

Check the apex, wildcard, and public TCP ports 80 and 443 before troubleshooting Caddy.

For backups, upgrades, security rules, and advanced diagnostics, see [docs/backup-restore.md](#), [docs/upgrading.md](#), [docs/security.md](#), and [docs/vps-installation.md](#).

You now have a local-first ESP registered in Device Manager and remotely accessible over HTTPS.

Arduino and OTA will be added in a later edition.